Varun Sai Machabattula

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ABOUT ME

Techno-economic specialist (M.Sc., RWTH Aachen) with 3 years of experience across business development, project controls and strategy in battery recycling, power generation and chemicals. Experienced in CAPEX/OPEX modelling, business cases, project scheduling and Power BI reporting, working at the intersection of Technology and Management.

WORK EXPERIENCE

Business Development & Sales Intern — **Aug 2026 – Present**
Primobius GmbH · EV & Consumer battery recycling — *Mönchengladbach, Germany*

- Built the CAPEX cost model (€150 M) that underpins budget decisions for new battery recycling projects.
- Built the WBS along the full recycling process flow and co-developed the resource-loaded project schedule.
- Sourced 10+ supplier quotations, performed product gap analysis and fed vendor pricing into the financial calculations.
- Analyse battery recycling market trends and competitors to set business development and sales priorities.
- Prepare management reports and presentations on market developments and project economics.

Working Student – Asset Integration & Strategy — **Apr 2025 – Apr 2026**
Uniper Kraftwerke GmbH — *Düsseldorf, Germany*

- Evaluated OPEX assumptions and business-case inputs for 3 new-build CCGT, hydrogen and CCS projects and a BESS project.
- Ran cost-escalation, sensitivity and vendor-assumption analyses to compare long-term supplier and service proposals.
- Built Power BI and Excel dashboards tracking project pipeline, milestones and risks, used in strategy reviews.
- Developed maintenance-strategy and outage-planning inputs for the 15-year lifecycle of a carbon capture plant.
- Coordinated with technology providers and internal teams to collect project inputs and clarify requirements.

Project Administrator — **Nov 2021 – Jul 2023**
Eastman Chemical Company · Hyderabad Engineering Centre — *Hyderabad, India*

- Tracked schedules, costs and risks across 3 concurrent engineering projects, flagging deviations against milestones.
- Built a Power Apps and Power Automate request system on SharePoint that replaced email-based engineering requests from the US headquarters, auto-assigned work across engineering teams, and tracked estimated vs. actual spend in Power BI.
- Built 5+ Power BI KPI dashboards that exposed schedule bottlenecks, improving project health tracking capabilities.
- Prepared monthly performance reports and leadership presentations, linking engineering, procurement and operations.
- Standardised reporting templates, structured data workflows in SharePoint and maintained a lessons-learned database.

SELECTED PROJECTS

Master Thesis – Investigating Green Hydrogen Innovation Ecosystems · Grade 1.7 — **Sep 2025 – Mar 2026**
RWTH Aachen

- Interviewed 10 experts from production to end use to map collaboration in a regional green hydrogen value chain.
- Identified coordination barriers and success factors; derived recommendations.

Strategic Technology Management – Blockchain for Smart Grids — **Apr 2024 – Jul 2024**
RWTH Aachen

- Assessed the market opportunity for integrating blockchain into sustainable smart-grid solutions.
- Developed a strategic-partnership and technology-diffusion plan to bring the solution to market.

ASME Human Powered Vehicle Challenge – Team Lead — **Sep 2017 – Mar 2019**
Osmania University

- Led a 25-member team to design and build a recumbent human-powered vehicle within budget; finished 2nd overall (2018).

EDUCATION

RWTH Aachen University — **Oct 2023 – Mar 2026**
M.Sc. Management and Engineering in Technology, Innovation, Marketing & Entrepreneurship · Grade 1.9

Osmania University, Hyderabad, India — **Aug 2016 – Sep 2020**
B.E. Mechanical Engineering

SKILLS

Project controls	WBS, resource-loaded scheduling, cost and milestone tracking, risk reporting, KPI dashboards
Commercial	CAPEX/OPEX models, business cases, sensitivity analysis, vendor evaluation, market and gap analysis
Tools	Excel (advanced, incl. VBA), Power BI, PowerPoint, Python, SQL, Power Apps
Languages	English (C1), German (B1)